Historical Species Analysis Report

ALA Datasets Analysis

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Executive Summary

Data overview:

Total Records: 22318

Total Species: 1269

Date Range: 1886-2024(138 years)

GPS Records: 22318

Unique Locations: 4894

Temporal Pattern:

Recent Records (last 5 years): 4429,

Historical Records: 17889

Recent Species: 699

Historical Species: 1099

Key finding

Species missing since 2000: 49

Species with recent records :499

Survey coverage : Good

TIMELINE ANALYSIS INTERPRETATION

Key Survey Patterns:

• Survey span: 1886 – 2024 (138 years)

• Peak activity: 2009 (4,862 records)

• Average annual records: 310

• Recent trend: ↗ Increasing

Survey Periods:

• Early era (1886–1950): Limited, opportunistic surveys

• Development era (1950–2000): Systematic biological surveys

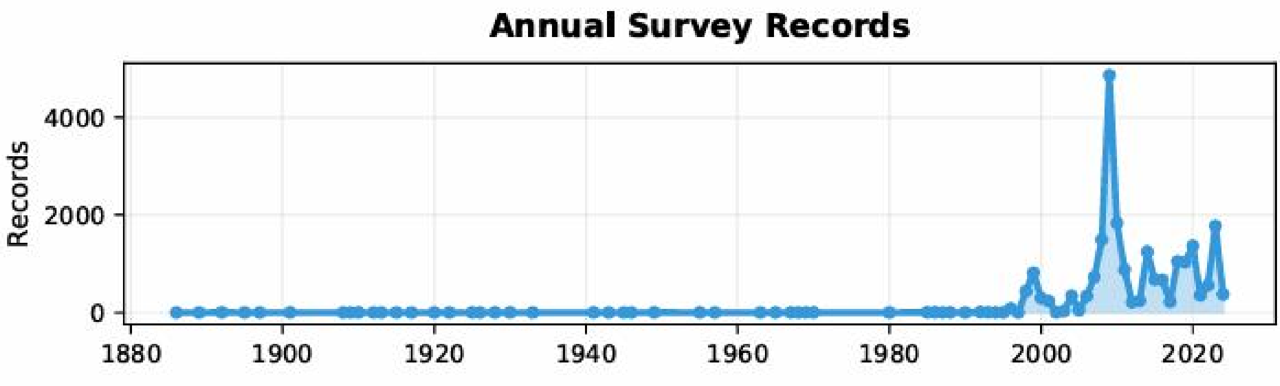
• Modern era (2000–2024): Technology-enhanced, intensive monitoring

Survey Quality Assessment:

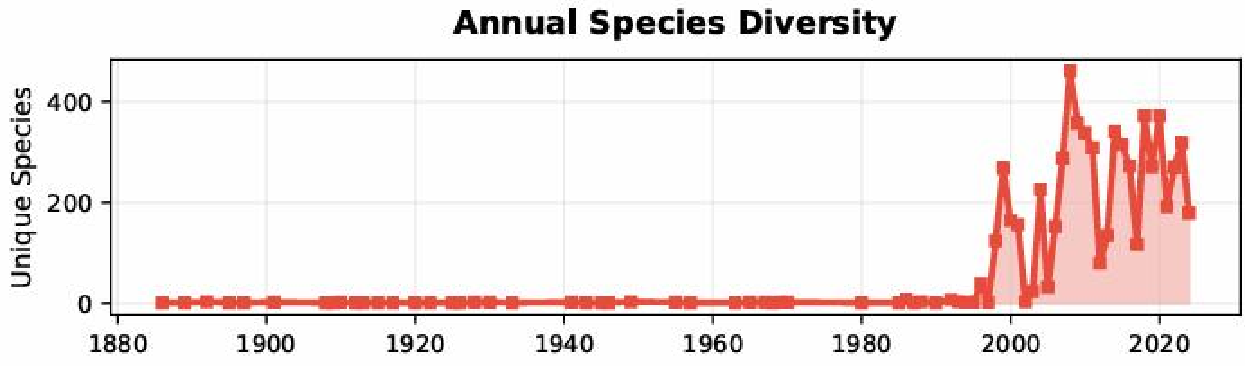
• Data quality: Variable

• Coverage consistency: Good

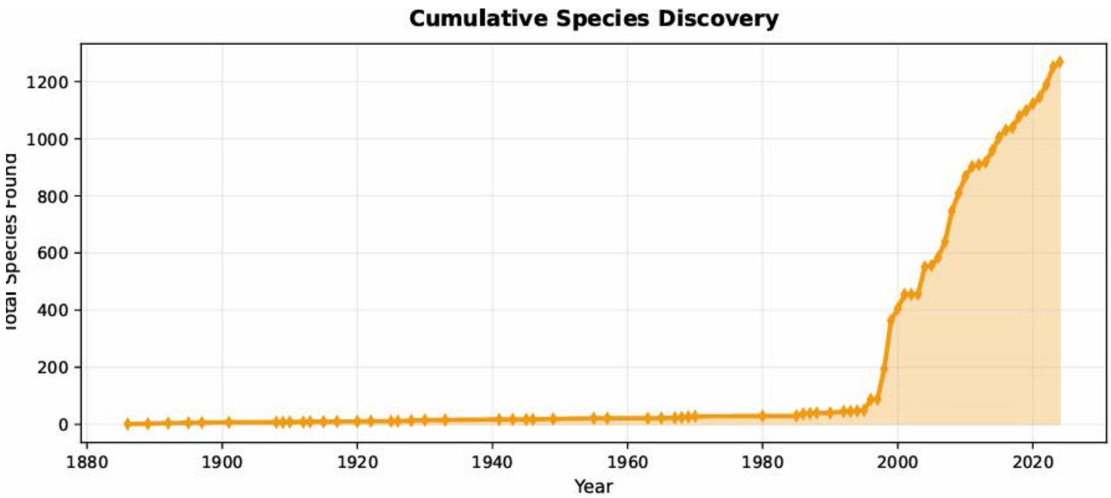
• Recent activity: Active



Survey records remained sparse before 1950, with very limited data. After 2000, record numbers increased sharply, peaking in 2008
with nearly 5,000 entries. This reflects intensive, large-scale survey efforts in recent decades



The number of unique species recorded per year was generally below 50 before 2000. From 2000 onwards, species diversity rose
dramatically, often exceeding 300 species per year, indicating broader and more systematic survey coverage.



The cumulative discovery curve is nearly flat before 1950, then grows gradually. After 2000 the curve rises steeply, surpassing 1,200
species by 2024. The sustained upward trajectory indicates ongoing discoveries and improving survey intensity.

Missing Species Analysis

Critical Findings:

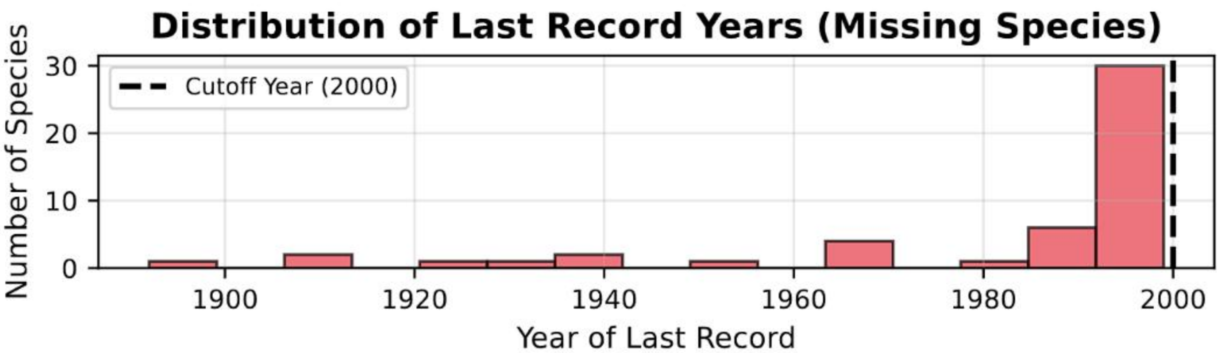
- 49 species (3.9% of total) have no records since 2000
- Oldest missing species: Tetratheca juncea (last seen 1892)
- Most recently missing: Pultenaea parviflora (last seen 1999)
- Average years missing: 43 years

Conservation Implications:

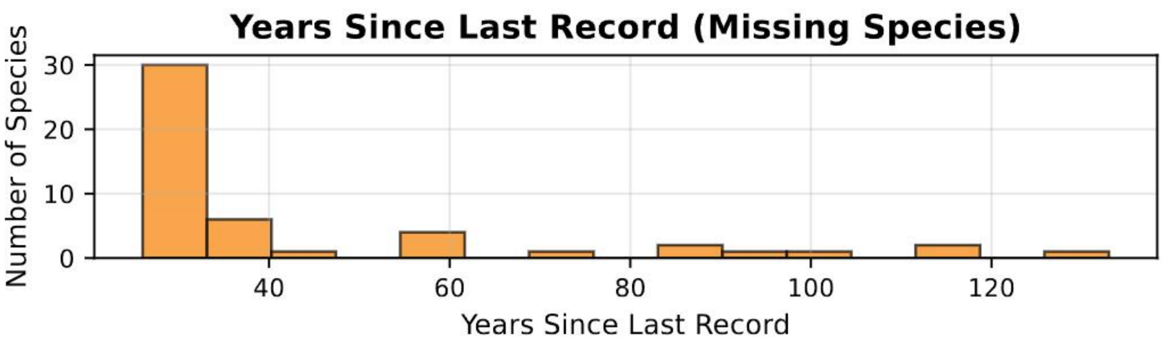
- These species may be locally extinct, extremely rare, or under-surveyed
- Priority should be given to species missing >50 years
- Historical habitat locations should be re-surveyed
- Some species may have been misidentified or taxonomically revised

Recommended Actions:

- Targeted surveys in historical locations
- Literature review for taxonomic changes
- Habitat assessment for ecological suitability
- Expert consultation for identification verification



Most species without records since 2000 were last observed in the late 20th century (1980s 1990s), with a strong pile-up near the cutoff year. A small number have not been seen for >100 years. This pattern suggests many taxa may still persist but were overlooked as effort shifted, making historical localities high-priority resurvey targets



The distribution is strongly right-skewed: most missing species fall in the 25 40-year band, with a long tail extending to 60 130 years.

A practical triage is to treat >50 years as high-priority resurvey candidates, 30 50 years as medium priority, and <30 years as lower priority.

Spatial Survey Coverage Analysis

SPATIAL COVERAGE INTERPRETATION

Geographic Scope:

Latitude range: -33.9820° to -33.8740° (0.1080° span)

Longitude range: 150.9667° to 151.1540° (0.1872° span)

Total survey area: Approximately 249 km²

Survey Distribution:

Historical records: 17,889 GPS points

Recent records: 4,429 GPS points

Coverage shift: Recent surveys cover new areas

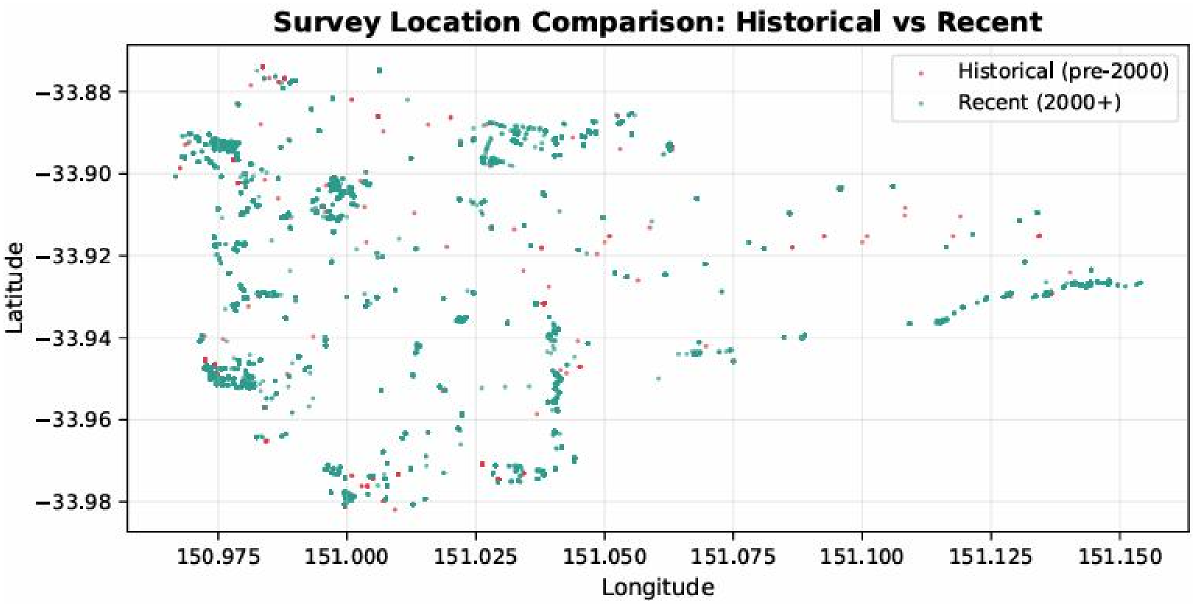
Key Observations:

Survey intensity varies significantly across the landscape

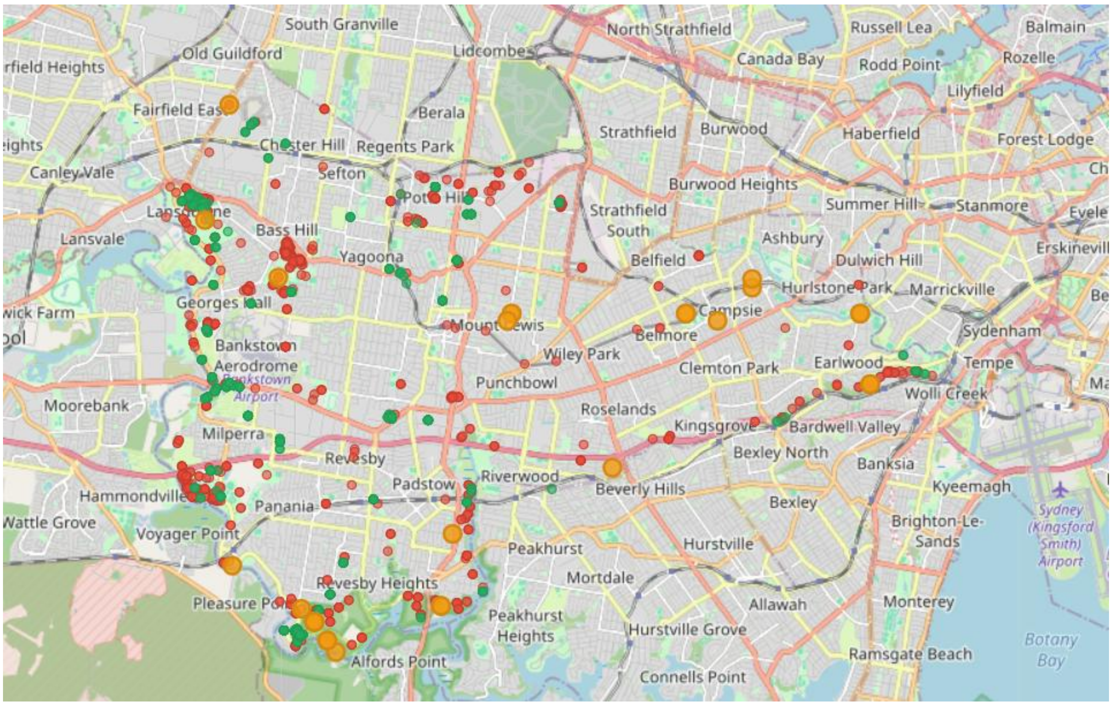
Some areas show clustering of survey points (intensive study sites)

Other areas show sparse coverage (potential survey gaps)

Recent surveys do cover similar areas to historical surveys



Recent surveys cover the landscape far more densely than historical efforts, with points clustering in several latitudinal bands that likely follow roads or fixed transects. Many historical localities have been re-visited, indicating good temporal overlap for trend comparisons, yet large gaps remain between bands and toward the eastern edge of the area. These under-sampled zones suggest an accessibility bias and may conceal additional species or range limits. Priority effort should target grids that are “historical-only” or currently blank, while capping repeat visits along the most intensively sampled bands.



📊 Map Legend

- Historical Survey Locations (before 2020)
- Recent Survey Locations (2020+)
- Priority Areas (Missing Species Locations)

[Species Survey Locations]

Reserve Survey Uniformity Analysis

RESERVE UNIFORMITY ASSESSMENT

Survey Evenness Analysis:

Reserves analyzed: 30

Average temporal coverage: 0.46 (1.0 = perfect coverage)

Average spatial uniformity: 0.00 (1.0 = perfectly even)

Uniformity Findings:

Most reserves show uneven survey distribution

Temporal gaps are common (surveys cluster in certain years)

Spatial clustering indicates preferred survey locations

Some areas within reserves are never surveyed

Problem Areas:

Reserves with temporal coverage <0.3: Need regular monitoring

Reserves with spatial uniformity <0.1: Need broader area coverage

Large reserves often have significant internal gaps

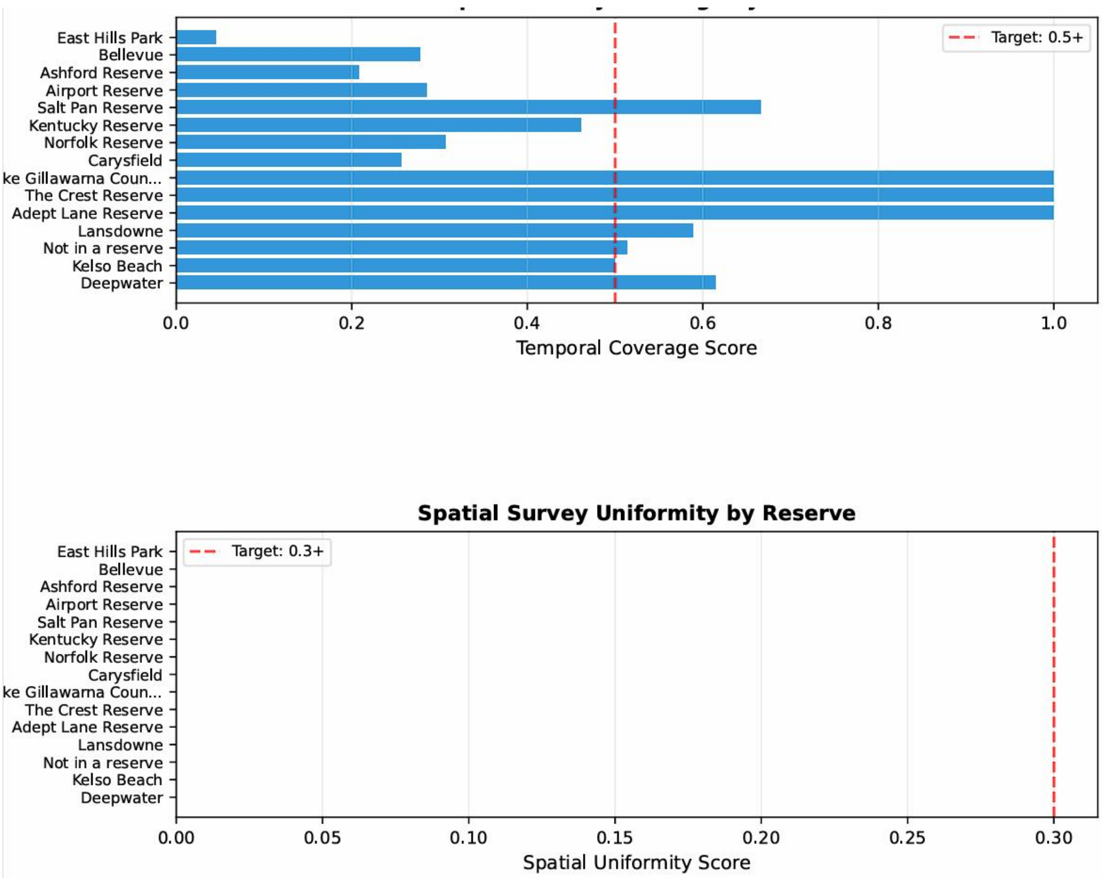
Recommended Actions:

Implement systematic grid-based survey protocols

Schedule regular monitoring across all reserve areas

Focus on previously unsurveyed sections within reserves

Establish permanent monitoringots for long-term studies



🌿 Species Observation Records

Scientific Name	Last Year	GPS Locations	Total Records	Years Missing
Tetratheca juncea	1892	3	3	**133** ⚠
Gonocarpus longifoliu	1910	2	2	**115** ⚠
Acacia bynoeana	1913	1	1	**112** ⚠
Lyperanthus suaveole	1922	1	1	**103** ⚠
Deyeuxia appressa	1930	1	1	**95** ⚠
Polyosma rigidiuscula	1941	1	1	**84** ⚠
Litsea leefeana	1941	1	1	**84** ⚠
Pterostylis acuminata	1955	1	1	**70** ⚠
Centipeda minima	1968	1	1	**57** ⚠
Philotheca obovalis	1969	1	1	**56** ⚠
Austrostipa elegantiss	1970	1	1	**55** ⚠
Acacia burkittii	1970	1	1	**55** ⚠
Ptilotus seminudus	1980	1	1	45 ⚡
Drosera auriculata	1986	1	1	39 ⚡
Drosera glanduligera	1986	1	1	39 ⚡
Utricularia uliginosa	1986	1	1	39 ⚡
Utricularia australis	1986	1	1	39 ⚡
Utricularia dichotoma	1986	1	1	39 ⚡
Utricularia lateriflora	1986	1	1	39 ⚡
Cenchrus longisetus	1992	2	2	33 ⚡
Anisomeles moschata	1993	8	8	32 ⚡
Chloris barbata	1995	1	1	30 ⚡
Epacris purpurascens	1995	1	1	30 ⚡
Chorizema parviflorur	1996	1	1	29 ⚡
Acacia echinula	1996	1	1	29 ⚡
Senecio diaschides	1996	1	1	29 ⚡
Acacia maidenii	1996	1	1	29 ⚡
Pultenaea mollis	1996	1	1	29 ⚡
Parsonsia lanceolata	1998	2	2	27 ⚡
Wurmbea dioica subsp	1998	2	2	27 ⚡
Sarcopetalum harveya	1998	2	2	27 ⚡
Melicope micrococca	1998	2	2	27 ⚡
Mentha diemenica	1998	2	2	27 ⚡